HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented MES recipes in **Emerson DeltaV MES (Syncade)**, translating business requirements into configurable electronic batch records for pharmaceutical API manufacturing.
- Authored and maintained high-quality, well-documented MES recipe configurations, ensuring ease of maintenance and adherence to quality standards for a biotech manufacturing facility.
- Collaborated with site teams to configure and support interfaces between the MES system and other enterprise applications, including **ERP**, automation controls, and lab systems, ensuring seamless data flow.
- Investigated and troubleshot production issues within the MES environment, determining root causes and implementing solutions or recommendations for system improvements, reducing critical incidents by **15%**.
- Generated computer system validation documentation in accordance with site procedures and **GMP** guidelines, ensuring compliance with regulatory standards for pharmaceutical production.
- Developed and executed test cases for new MES recipes and configuration changes, performing system and integration testing to validate functionality against business requirements and quality standards.
- Containerized MES recipe deployment packages using **Docker**, facilitating consistent and repeatable deployments across multiple environments within the AWS cloud infrastructure.

SKILLS

MES Platform – Emerson DeltaV MES (Syncade), MES Recipe Authoring, Electronic Batch Records, Manufacturing Execution Systems

Programming & Scripting – Visual Basic, Python, SQL, Shell Scripting

Core Technologies – MS SQL Server, AWS, Docker, Linux, Git, Jenkins

Methodologies & Practices – Agile, Scrum, Change Management, Configuration Management, GMP, Quality Standards, System Validation

Domain Knowledge – Biotech Manufacturing, Pharmaceutical API Production, IT Application Support, ERP Integration, Automation Systems, Lab Systems

PROJECTS

MES Recipe Authoring and Validation (HCL Technologies)

- Developed and configured complex MES recipes for a pharmaceutical client's API manufacturing line, incorporating specific process steps, material controls, and operator instructions.
- Authored detailed design documentation and validation test scripts for new MES recipes, ensuring alignment with business requirements and compliance with **GMP** regulations.
- Executed system and integration testing for MES recipe changes, documenting results and collaborating with business analysts and operations teams to achieve UAT sign-off.

Automated Data Reporting for Manufacturing Operations (HCL Technologies)

- Engineered Python scripts to extract and transform data from operational databases, generating daily/weekly summary reports for manufacturing performance metrics.
- Developed SQL queries against **MS SQL Server** to aggregate data for reporting dashboards, improving data accuracy and reducing manual reporting effort by approximately **40%**.
- Managed code versions and deployments using **Git** and integrated scripts into scheduled jobs on a Linux environment for automated execution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)